DEGREE MINIMALITY IN THE EQUIVARIANT CLASS OF THE ALPÖGE KELLER MAP, AND THE MOMENT-MAP STRUCTURE OF ITS COTANGENT LIFT

DANIEL KIRTCHAKOV

Abstract. Let $F\colon \mathbb{C}^3 \to \mathbb{C}^3$ be the degree-7 Keller map announced by L. Alpöge on July 19, 2026, which refutes the Jacobian conjecture in dimension three. Our two new results concern its $\mathbb{C}^\times$-equivariant class, in which $F = (A/x^2, B/x, xC)$ with A, B, C polynomials in the invariants $u = 1 + xy$, $s = x^2z$.

First, a *degree-minimality theorem*: in that class no Keller map of degree ≤ 6 exists at all in the sector where C genuinely involves s — the sector containing Alpöge's map — and every Keller map of degree ≤ 6 elsewhere in the class is a polynomial automorphism of $\mathbb{C}^3$. Degree 7 is therefore minimal in the class. The emptiness statements rest on eight Gröbner-basis unit-ideal certificates computed over $\mathbb{Q}$ (msolve, multi-modular F4) and reproduced mod 32003, together with a *no-go lemma* valid for every weight $k \geq 1$ and every degree: if C is s-free and A, B are at most affine in s, the map is an automorphism.

Second, a *moment-map identity*. The $\mathbb{C}^\times$-action lifts Hamiltonianly to $T^*\mathbb{C}^3$, and the cotangent lift $\Phi(q, p) = \big(F(q), (JF(q))^{-\intercal}p\big)$ preserves the moment map exactly: $\nu \circ \Phi = \mu$ with $\mu = xp_1 - yp_2 - 2zp_3$.

The map Φ itself is not ours: it was written down, and its exact preservation of the symplectic form, $\det J\Phi = 1$, and non-injectivity verified, by W. G. P. Mayner on July 21, 2026. We identify it as an explicit witness for the Poisson conjecture PC_3 via the Adjamagbo–van den Essen chain, record its component degrees $(7, 6, 4, 9, 10, 12)$ and an explicit rational triple collision in $\mathbb{C}^6$, and prove the quantization identity $\operatorname{gr}\Psi_F = \Phi^*$ for Mayner's Weyl-algebra endomorphism Ψ_F. The remaining structural material — the master equation, the base-point (anchor) identity, the S_3 cover, and the exact image theorem — was obtained independently here but is anticipated by Shaska, by the anonymous ulam.ai note, and by Mayner, all in July 2026; it is retained for self-containedness and priority is credited to those sources in Section 1.4. Everything is in characteristic zero; JC_2, DC_1, and DC_2 remain open.

1. Introduction

1.1. **The three conjectures.** Throughout, we work over $\mathbb{C}$; every statement in this note is in characteristic zero. For a polynomial map $F = (F_1, \ldots, F_n)\colon \mathbb{C}^n \to \mathbb{C}^n$ write $JF = (\partial F_i/\partial x_j)$ for its Jacobian matrix. F is a *Keller map* if $\det JF$ is a nonzero constant [Kel39]. Three long-standing conjectures assert that natural classes of endomorphisms are automorphisms:

- JC_n (Jacobian conjecture): every Keller map $\mathbb{C}^n \to \mathbb{C}^n$ is a polynomial automorphism.

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All computations in this note were carried out with Claude (Fable 5) driving SymPy 1.14 and msolve 0.10.1. Every displayed identity is asserted by a script listed in Section 7; nothing is conjectural unless labeled so. *Draft of August 4, 2026 — not for circulation.*

- DC_n (Dixmier conjecture [Dix68]): every unital $\mathbb{C}$-algebra endomorphism of the n-th Weyl algebra W_n is an automorphism.
- PC_n (Poisson conjecture): every unital $\mathbb{C}$-algebra endomorphism of the polynomial Poisson algebra $P_n = \big(\mathbb{C}[q_1, \dots, q_n, p_1, \dots, p_n], \{\,,\}\big)$ that preserves the Poisson bracket is an automorphism.

These are linked by the chain

$$\mathrm{JC}_{2n} \implies \mathrm{PC}_n \implies \mathrm{DC}_n \implies \mathrm{JC}_n, \tag{1}$$

established by Tsuchimoto [Tsu05], Belov-Kanel–Kontsevich [BKK07], Bavula [Bav05], and Adjamagbo–van den Essen [AvdE07] (who inserted the Poisson link); for the implication $\mathrm{DC}_n \Rightarrow \mathrm{JC}_n$ see [vdE00] and [BKK07, p. 2].

On July 19, 2026, Levent Alpöge announced an explicit counterexample to JC_3 [Alp26] — the map F of (2) below — thereby refuting JC_n for all $n \geq 3$. (The problem had been suggested to him by Akhil Mathew, and, per the announcement, the search that produced the map was run with Claude Fable 5. We do not describe the discovery process beyond the announcement; see Tao's expository post [Tao26] and the discussion at the Secret Blogging Seminar [Spe26].) By the chain (1), read contrapositively, the failure of JC_3 formally entails the failure of DC_3 and hence of PC_3. Both consequences are already on the public record: Mayner [May26] gave the explicit Weyl-algebra witness for $\neg\mathrm{DC}_3$ on July 20–21, 2026, and the Wikipedia article on the Jacobian conjecture states flatly that "the Dixmier and Poisson conjectures are false in every dimension $n > 2$", citing [AvdE07]. Nothing in Sections 3–4 should be read as a priority claim on either statement; what those sections contribute is an explicit PC_3-witness, its degrees and momenta, and the quantization identity relating it to Ψ_F. The new results of this note are the degree-minimality theorem (Theorem E) and the moment-map identity (Theorem C(i)); see Section 1.4 for a claim-by-claim priority table.

1.2. **The map.** Alpöge's map is $F = (F_1, F_2, F_3)\colon \mathbb{C}^3 \to \mathbb{C}^3$,

$$\begin{aligned} F_1 &= (1+xy)^3 z + y^2(1+xy)(4+3xy), && \deg F_1 = 7,\\ F_2 &= y + 3x(1+xy)^2 z + 3xy^2(4+3xy), && \deg F_2 = 6,\\ F_3 &= 2x - 3x^2 y - x^3 z, && \deg F_3 = 4, \end{aligned} \tag{2}$$

with $\det JF \equiv -2$. It is non-injective in a way anyone can check by hand in a minute:

$$F\Big(1, -\tfrac{3}{2}, \tfrac{13}{2}\Big) = F\Big(-1, \tfrac{3}{2}, \tfrac{13}{2}\Big) = F\Big(0, 0, -\tfrac{1}{4}\Big) = \Big(-\tfrac{1}{4}, 0, 0\Big), \tag{3}$$

a triple collision at rational points.

1.3. **Statement of results.** *What is new here is Theorem E and Theorem C(i)*, together with the no-go Lemma 6.3 on which Theorem E rests. Theorem E says that in the $\mathbb{C}^\times$-equivariant class of Alpöge's map, degree 7 is minimal: below it the relevant sector is *empty*, and what survives elsewhere in the class is automorphisms only. Theorem C(i) says that the $\mathbb{C}^\times$-action lifts Hamiltonianly to the cotangent bundle and that the cotangent lift preserves the moment map on the nose. The remaining theorems are stated because the note is meant to be self-contained; each carries its priority attribution in its own statement, and Section 1.4 gives the full table. The reader interested only in the new material may go directly to Sections 5.2 and 6.

Let $N = (JF)^{-1} = \operatorname{adj}(JF)/(-2)$, a 3×3 matrix with *polynomial* entries (of total degree at most 11, at most 14 monomials each — 77 in total across the nine entries), and let $D_j = \sum_{k=1}^{3} N_{kj}\, \partial_k$ for $j = 1, 2, 3$.

Theorem A (Dixmier conjecture fails for $n \geq 3$; *Mayner's theorem*, independently verified here)**.** *The assignments $x_i \mapsto F_i$, $\partial_j \mapsto D_j$ extend to a well-defined unital $\mathbb{C}$-algebra endomorphism $\Psi_F : W_3 \to W_3$. It is injective, and it is not surjective: the generator x is not in its image. Consequently DC_n is false for every $n \geq 3$, with explicit witness $\Psi_F \otimes \mathrm{id}_{W_{n-3}}$. This is due to W. G. P. Mayner (comment of July 20, 2026 on* [Spe26]*; note and report of July 21, 2026* [May26]*, where the non-member exhibited is y rather than x). We claim no priority whatsoever for it; Section 3 is an independent re-verification, retained because Theorem B and Theorem C(ii) are stated relative to Ψ_F.*

Theorem B (Mayner's cotangent lift, identified as a counterexample to the Poisson conjecture)**.** *The map Φ below and its three displayed properties are Mayner's* [May26, §8]*, who verified there that the cotangent lift $\Phi(q, p) = (F(q), G(q)p)$, $G = (JF)^{-\mathsf{T}}$, is a polynomial self-map of $\mathbb{C}^6$ which "preserves the standard symplectic form exactly", has $\det J\Phi = 1$ identically, and is not injective; his priority section lists "the symplectic lift to $\mathbb{C}^6$" among the items apparently not previously public. What is added here is: the identification of Φ^* as an explicit witness for PC_n through the Adjamagbo–van den Essen formulation* [AvdE07]*; the component degrees; the explicit momenta of a rational triple collision; and, in Theorem C, the moment map and the quantization identity. Precisely:*

$$\Phi : \mathbb{C}^6 \to \mathbb{C}^6, \qquad \Phi(q, p) = \big(F(q),\, (JF(q))^{-\mathsf{T}} p\big)$$

is a polynomial map with component degrees $(7, 6, 4, 9, 10, 12)$ whose Jacobian $M = J\Phi$ satisfies $M^{\mathsf{T}}\Omega M = \Omega$ identically and $\det M = 1$, where Ω is the standard symplectic matrix. Φ is generically 3:1; in particular the three rational points

$$a_1 = \Big(1, -\tfrac{3}{2}, \tfrac{13}{2};\, -\tfrac{405}{16}, \tfrac{25}{8}, -\tfrac{13}{8}\Big), \quad a_2 = \Big(-1, \tfrac{3}{2}, \tfrac{13}{2};\, -\tfrac{387}{16}, \tfrac{31}{8}, \tfrac{11}{8}\Big), \quad a_3 = \Big(0, 0, -\tfrac{1}{4};\, \tfrac{9}{2}, 2, 1\Big)$$

satisfy $\Phi(a_1) = \Phi(a_2) = \Phi(a_3) = \big(-\tfrac{1}{4}, 0, 0;\, 1, 2, 3\big)$. Consequently the pullback Φ^* is an injective, non-surjective Poisson endomorphism of P_3, and PC_n is false for every $n \geq 3$.

Theorem C (Equivariant structure; *part (i) is new*)**.**

(i) *The $\mathbb{C}^\times$-action $t\cdot(x, y, z) = (tx, t^{-1}y, t^{-2}z)$ lifts to a Hamiltonian action on $T^*\mathbb{C}^3 \cong \mathbb{C}^6$ with moment map $\mu = xp_1 - yp_2 - 2zp_3$; the target carries the action of weights $(-2, -1, 1)$ with moment map $\nu = -2Q_1P_1 - Q_2P_2 + Q_3P_3$. The map Φ is equivariant and preserves the moment map exactly: $\nu \circ \Phi = \mu$.*

(ii) *Ψ_F preserves the order filtration of W_3, and under the canonical isomorphism $\mathrm{gr}\, W_3 \cong \mathcal{O}(T^*\mathbb{C}^3)$ one has $\mathrm{gr}\, \Psi_F = \Phi^*$. In this precise sense Ψ_F is the canonical quantization of Φ, and quantization does not restore invertibility. (The symbol computation behind this — that the $\sigma_1(D_j) = \sum_k N_{kj}\xi_k$ form a $\mathbb{C}[x, y, z]$-basis of the degree-one part of $\mathrm{gr}\, W_3$ — is step 2 of Mayner's self-contained proof* [May26, §8]*; the identification of the resulting graded endomorphism with Φ^* is our packaging.)*

The $\mathbb{C}^\times$-equivariance of F itself is not new: it is recorded by Speyer [Spe26]*, in* [MO26]*, by Mayner* [May26, §6]*, and is the organizing hypothesis of Shaska* [Sha26]*. Part (i) — the Hamiltonian lift and the exact identity $\nu \circ \Phi = \mu$ — is, to our knowledge, not in any of them; we found no occurrence of "moment map" anywhere in the July–August 2026 literature on this example.*

Theorem D (Exact image; *anticipated*, see the attribution below)**.** *The image of F is $\mathbb{C}^3$ minus the punctured rational curve*

$$\Gamma = \Big\{ \Big(\tfrac{4}{27t^2},\, \tfrac{4}{3t},\, t\Big) : t \in \mathbb{C}^\times \Big\},$$

and consequently $\operatorname{im}\Phi = \mathbb{C}^6 \setminus (\Gamma \times \mathbb{C}^3)$. *Generic fibers of* F *have three points; the count drops (generically) to two over the pullback of* $\{\Delta_2 = 0\}$ *and to one over the pullback of* $\{\Delta_1 = 0\}$ *— two sheets escaping to infinity — and to zero exactly on* Γ*, where all three sheets have escaped. (*Δ_1, Δ_2 *are the explicit discriminant polynomials of Proposition 5.3.)*

Priority. The fiber counts 3/1/0 *and the identification of the missed locus with the curve* Γ *were obtained first in the anonymous note* [Ula26, Thm. 4.2] *(PDF timestamp July 20, 2026), and independently in* [May26, §4.3] *and in* [Sha26, Prop. 5.1]*, where the same curve appears in the same normalization,* $\{(\frac{4}{27}t^{-2}, \frac{4}{3}t^{-1}, t)\}$. *Our proof is independent and is retained only for self-containedness and because the transfer to* $\operatorname{im}\Phi$ *is used later.*

Theorem E (Degree minimality in the equivariant class; *new*)**.** *In the equivariant class of weights* $(1, -1, -2)$ *— maps* $F = (A/x^2,\ B/x,\ xC)$ *with* $A, B, C \in \mathbb{C}[u, s]$*,* $u = 1+xy$*,* $s = x^2z$*, subject to the polynomial-lift conditions — every Keller map of degree* ≤ 6 *is a polynomial automorphism of* $\mathbb{C}^3$. *Stronger: in the entire sector where* C *genuinely involves* s *(the sector containing Alpöge's map), no Keller map of degree* ≤ 6 *exists at all, automorphisms included. Hence degree* 7 *is minimal in this class. Unconditional minimality of degree* 7 *among all counterexamples in* $\mathbb{C}^3$ *remains open.*

Theorem E is proved in Section 6; it rests on the no-go Lemma 6.3 (valid for all weights $k \geq 1$ and all degrees, and also new) together with eight Gröbner-basis unit-ideal certificates computed over $\mathbb{Q}$ (msolve, multi-modular F4) and reproduced mod 32003. We record how it sits against the nearest published statements.

- Shaska [Sha26, Thm. 10.10] proves that no graded Keller counterexample in dimension three has *both* A and B of degree one in the invariants, for any signature (r, s) — a strictly weaker emptiness statement, and one he explicitly leaves incomplete: "the smallest cases left open in dimension three are $(r, s) = (1, 1)$ with $\mathbf{d} = (2, 1, d_3)$ and $\mathbf{d} = (1, 2, d_3)$; whether $\mathcal{K}(3, (1, -1, -1))$ is empty we do not know." Theorem E clears an entire degree band, in his hyperbolic signature $(r, s) = (1, 2)$, with no degree-one hypothesis on A or B.
- Mayner [May26, §7] proves a *quadratic no-go* inside a different ansatz — the z-linear line congruences with a quadratic direction field — concluding that "within the parallel-congruence construction, degree 3 is the minimum degree of the covering". That is a statement about the covering degree in his structural class; Theorem E is a statement about the total degree in the $\mathbb{C}^\times$-equivariant class. Neither contains the other.
- Jelonek [Jel26] shows that if $X(n, d)$ is irreducible then for $n \geq 3$, $d \geq 6$ a *generic* element is a counterexample. That is the opposite direction: an abundance statement above degree 6, not an emptiness statement below it, and it is conditional on irreducibility.
- Unconditionally, invertibility of Keller maps is known only through degree 2, in every dimension, by Wang's theorem [Wan80]; and ulam.ai [Ula26, Cor. 5.3] exhibits non-proper Keller maps of every generic degree $d \geq 3$ (their total degrees grow, and none of their degree-≤ 6 members is equivariant of our type). Theorem E is therefore the first emptiness result we know of that rules out an explicit degree band inside a class known to contain a counterexample.

The remaining proofs occupy Sections 3–5. Every displayed identity is asserted by a script listed in Section 7.

1.4. **Provenance and credit.** This subsection was rewritten on August 4, 2026 after a first-hand check of the prior art, in which several results stated here turned out to have earlier public sources. What follows is the corrected record.

The map. The counterexample F is Alpöge's, announced July 19, 2026; the problem was posed by Akhil Mathew, and the search was run with Claude Fable 5, per the announcement. Everything in this note is downstream of that example.

The Weyl endomorphism <u>*and*</u> *the symplectic lift.* Both are W. G. P. Mayner's, from the same day and the same source. The explicit operators D_j and the statement that Ψ_F refutes DC_3 were made public in a comment of July 20, 2026 on the Secret Blogging Seminar thread [Spe26] and in a six-page informal note of July 21, 2026 (prepared with Claude Fable 5); the accompanying report, in the same repository and the same two commits of July 21, 2026, contains in §8 the cotangent lift $\Phi(q,p) = (F(q), G(q)p)$ together with the statement — machine-verified there — that it preserves the standard symplectic form exactly, that $\det J\Phi = 1$, and that it is not injective; his verified-facts table lists it as item 24, and his priority section §12 lists "the symplectic lift to $\mathbb{C}^6$" as item 3 among the results apparently not previously public [May26]. Mayner also records the correct caution that this does *not* bear on the Belov-Kanel–Kontsevich automorphism conjecture, a caution we repeat in Section 1.5. An earlier draft of the present note credited Mayner for the Weyl endomorphism only; that was an error of attribution, and it is corrected here. Mayner's note additionally goes further than our Section 3: it computes the exact image $\bigoplus_\alpha \mathbb{C}[F]D^\alpha$, exhibits non-members, tabulates the codimension of the image in filtration degrees $d \le 7$, and observes that the cokernel is not finitely generated. We claim no priority for either Ψ_F or Φ.

Independently obtained, subsequently found to be anticipated. The material of Sections 5.1, 5.3, 5.4 and Lemma 6.1 was obtained here independently, before we became aware of the sources below; it is *not* new, and we retain it only for self-containedness. Priority belongs to those sources, as follows. All dates are 2026 and were checked against the sources themselves (arXiv submission records, PDF timestamps, Git commit timestamps, StackExchange creation timestamps) on August 4, 2026.

Item (as stated here)	Earliest public source we could verify	Date
Master equation, Prop. 5.1	Shaska [Sha26, Thm. 8.3] (identical trilinear bracket identity, $\Lambda[B, A] + sA[B, \Lambda] + rB[\Lambda, A] = \kappa$, with Keller $\iff$ constant); his §9.2 is titled "The master equation". Only in v2 — v1 has the divisorial form $\mathrm{Jac}_{u,v}(P, Q) = \kappa\Lambda^2$ as Thm. 6.1.	Jul 25
Parked square: $\det JG = \lambda C^k$, order-two vanishing on the contracted line	Shaska [Sha26, Thm. 7.1] (= Thm. 6.1 of v1, Jul 22)	Jul 22
Anchor lemma, Lemma 6.1	Shaska [Sha26, Lem. 8.5]: evaluating the master equation at the base point gives $\Lambda(O)\,[B, A](O) = \kappa$, which is our $\lambda = C_0(1)B_0'(1)A_1(1)$ after translating $u \mapsto u-1$.	Jul 25
S_3 monodromy, non-Galois, discriminant not a square	[MO26] (explicit cubic model, the discriminant, S_3, trivial deck group, and the Campbell/Razar/Wright "Galois case" theorem quoted for exactly the reason we quote it); Lou [Lou26]; Mayner [May26, §4.4]; Shaska [Sha26, Thm. 4.4(2)]	Jul 20
Trace identity (fiber cubic has no subleading term)	Mayner [May26, §4.2], in the coordinate x ("the x-coordinates of the preimages sum to zero"); Shaska [Sha26, Rem. 5.4], in the coordinate s (the roots sum to $2 = \kappa$). Our version is the same phenomenon in the invariant coordinate u.	Jul 21
Exact image, Theorem D	ulam.ai [Ula26, Thm. 4.2]; Mayner [May26, §4.3]; Shaska [Sha26, Prop. 5.1]	Jul 20

What is new here. Two things, plus one reframing.

(a) *Theorem E*, degree minimality in the equivariant class, with its eight unit-ideal certificates and the degree-7 positive control. We have found nothing comparable; the nearest statements are compared in Section 1.3.

(b) *Theorem C(i)*, the Hamiltonian lift and the exact moment-map identity $\nu \circ \Phi = \mu$; and *Lemma 6.3*, the no-go lemma for all weights $k \geq 1$ and all degrees.

(c) A *reframing*: Mayner's Φ is identified as an explicit PC_n-witness through the Adjamagbo–van den Essen formulation [AvdE07], and tied to Ψ_F by $\mathrm{gr}\,\Psi_F = \Phi^*$ (Theorem C(ii)). The object is not new and the conclusion $\neg\mathrm{PC}_n$ is not new — it follows formally from $\neg\mathrm{DC}_3$ and is already asserted in secondary sources. What is added is that the witness is written down, its degrees and momenta computed, and its non-invertibility checked in two lines from (3) with no filtration or reduction argument.

Minimality certificates for the other weights $k = 1$ and $k = 3$ are in preparation and are *not* claimed here.

Limits of this survey. The table above rests on a targeted search of the July–August 2026 record: the arXiv listings for math.AG and math.RA in that window, the Speyer and Tao threads, the two MathOverflow questions cited, the ulam.ai note, Lou's derivation, and Mayner's repository. A negative cannot be proved this way. In particular we were unable to retrieve MathOverflow question 513390, referenced in [May26, §12]; it returns a 404 and does not appear in the StackExchange API, so we do not cite it and record its content as unverified.

1.5. **What is not claimed.** JC_2 (the planar Jacobian conjecture), DC_1, and DC_2 remain open; for DC_1 there is a claimed proof by Zheglov [Zhe24] currently under review, and see Bavula–Levandovskyy [BL20] for prior partial results. The Belov-Kanel–Kontsevich conjecture $\mathrm{Aut}(W_n) \cong \mathrm{Aut}(P_n)$ [BKK05] is *not* refuted by anything here: our maps are proper endomorphisms, not automorphisms, so they simply do not belong to the groups in question. This caution is Mayner's, stated in the same place as the lift itself ([May26, §8]: "an earlier draft called this construction 'Kontsevich-relevant', which overstates the connection"), and we repeat it because it is easy to get wrong. The pair (Φ, Ψ_F) is, however, a sharp stress test for the proposed lifting machinery relating polynomial symplectomorphisms to Weyl-algebra automorphisms [KBEY18], whose status we do not assess here: any such machinery must be compatible with the existence of a non-invertible symplectic endomorphism whose canonical quantization is a non-invertible Weyl endomorphism. All statements are in characteristic zero.

2. Certified facts about F and its cotangent lift

We collect the machine-verified facts used throughout. Script names refer to Section 7.

Proposition 2.1. *For F as in (2):*

(i) $\det JF = -2$ *identically, so F is a Keller map and an everywhere-local biholomorphism.*

(ii) *The triple collision (3) holds; the three source points are distinct points of $\mathbb{Q}^3$. A further rational fiber point on the generic side: $F(-\frac{1}{4}, -4, 48) = (608, -232, 1)$.*

(iii) $N := (JF)^{-1} = \mathrm{adj}(JF)/(-2)$ *has polynomial entries with rational coefficients, of total degree at most* 11.

(Verified in `core_verify.py` and `weyl_verify.py`; the nine expanded entries of N are listed in `weyl_endomorphism.txt`.)

Two representative entries of N (columns index the operators D_j of Section 3):

$$N_{11} = 3x^6yz + 9x^5y^2 + 3x^5z + 3x^4y - 4x^3, \qquad N_{21} = 3x^4yz + 9x^3y^2 + 3x^3z + 3x^2y - 3x.$$

Definition 2.2 (Mayner [May26, §8])**.** The *cotangent lift* of F is $\Phi(q,p) = \big(F(q),\, (JF(q))^{-\mathsf{T}}p\big)$, a polynomial map $\mathbb{C}^6 \to \mathbb{C}^6$ by Proposition 2.1(iii). This map, and the facts recorded in Proposition 2.3 that it preserves the symplectic form exactly, has $\det J\Phi = 1$, and is non-injective, are due to Mayner; our verification is independent. We write $\Omega = \left(\begin{smallmatrix} 0 & I_3 \\ -I_3 & 0 \end{smallmatrix}\right)$ for the standard symplectic matrix in coordinates (x, y, z, p_1, p_2, p_3).

Proposition 2.3 (Mayner [May26, §8]; component degrees added here)**.** *$M := J\Phi$ satisfies $M^{\mathsf{T}}\Omega M = \Omega$ identically on $\mathbb{C}^6$, and $\det M = 1$ (as must hold for any symplectic matrix; it is verified independently). The component degrees of Φ are $(7, 6, 4, 9, 10, 12)$. In particular Φ has everywhere-invertible differential, hence open image, hence is dominant. (Verified in `dixmier_symplectic_verify.py`; the first two assertions are Mayner's, verified there by expansion of $\Phi^*\omega - \omega$ and of $\det D\Phi$.)*

Since a fiber of Φ over (Q, P) is $\{(q,\, JF(q)^{\mathsf{T}}P) : F(q) = Q\}$, the fibers of Φ are in canonical bijection with those of F: all fiber statements (generic 3:1, collisions, the missed locus) transfer verbatim between F and Φ. The three points a_i of Theorem B are exactly the collision fiber (3) decorated with momenta $p = JF(q_i)^{\mathsf{T}}(1, 2, 3)^{\mathsf{T}}$.

3. The Weyl endomorphism and the Dixmier conjecture

Let $W_3 = \mathbb{C}\langle x_1, x_2, x_3, \partial_1, \partial_2, \partial_3\rangle$ with $[\partial_i, x_j] = \delta_{ij}$, $[x_i, x_j] = [\partial_i, \partial_j] = 0$, and write $(x_1, x_2, x_3) = (x, y, z)$. Recall the *order filtration* $W_3^{(m)} = \operatorname{span}\{x^\beta \partial^\alpha : |\alpha| \le m\}$; its associated graded is the commutative polynomial algebra $\operatorname{gr} W_3 \cong \mathbb{C}[x, y, z, \xi_1, \xi_2, \xi_3] = \mathcal{O}(T^*\mathbb{C}^3)$ via the principal-symbol map, carrying the standard Poisson bracket. We identify (ξ_1, ξ_2, ξ_3) with the momenta (p_1, p_2, p_3) of Section 2.

Proposition 3.1 (CCR; machine-verified)**.** *With $D_j = \sum_k N_{kj}\partial_k$:*

(A) *$[D_j, F_i] = \delta_{ij}$ for all i, j; equivalently $JF \cdot N = I_3$, nine polynomial identities.*

(B) *$[D_i, D_j] = 0$ for all i, j; equivalently the nine coefficient polynomials $\sum_k \big(N_{ki}\,\partial_k N_{lj} - N_{kj}\,\partial_k N_{li}\big)$ vanish identically.*

Hence $x_i \mapsto F_i$, $\partial_j \mapsto D_j$ extends to a well-defined unital endomorphism $\Psi_F\colon W_3 \to W_3$. (Verified by `weyl_verify.py`; independently re-checked in `dixmier_symplectic_verify.py`.)

Identity (A) is a repackaging of $N = (JF)^{-1}$; the substantive content is (B), the pairwise commutativity of the vector fields D_j, which encodes precisely the flatness of the "inverse coordinate frame" of the non-invertible map F.

Lemma 3.2. *Ψ_F preserves the order filtration, and $\operatorname{gr}\Psi_F = \Phi^*$ as endomorphisms of $\mathcal{O}(T^*\mathbb{C}^3)$.*

Proof. $\Psi_F(x^\beta\partial^\alpha) = F^\beta D^\alpha$ has order $\le |\alpha|$, since each D_j has order one and the order of a product is at most the sum of the orders; hence $\Psi_F(W_3^{(m)}) \subseteq W_3^{(m)}$ and the graded endomorphism is defined. On generators, $\operatorname{gr}\Psi_F(x_i) = F_i$ and $\operatorname{gr}\Psi_F(\xi_j) = \sigma_1(D_j) = \sum_k N_{kj}\xi_k$. On the other side, writing $B = (JF)^{-\mathsf{T}} = N^{\mathsf{T}}$,

$$\Phi^*(\xi_j) = \big(Bp\big)_j = \sum_k B_{jk}\,\xi_k = \sum_k N_{kj}\,\xi_k, \qquad \Phi^*(x_i) = F_i.$$

The two algebra maps agree on generators, hence everywhere. □

Lemma 3.3. *$\operatorname{ord}\Psi_F(S) = \operatorname{ord} S$ for every nonzero $S \in W_3$. In particular every preimage of an order-zero operator has order zero.*

Proof. Φ is dominant (Proposition 2.3), so Φ^* is injective on $\mathcal{O}(\mathbb{C}^6)$. If $\operatorname{ord} S = m$ with principal symbol $\sigma_m(S) \ne 0$, then the class of $\Psi_F(S)$ in $W_3^{(m)}/W_3^{(m-1)}$ is $\Phi^*\sigma_m(S) \ne 0$ by Lemma 3.2, so $\operatorname{ord}\Psi_F(S) = m$. □

Proof of Theorem A. Well-definedness is Proposition 3.1. Injectivity is automatic: W_3 is a simple ring and $\Psi_F(1) = 1$, so $\ker\Psi_F$, a two-sided ideal not containing 1, is zero. For non-surjectivity, suppose $x = \Psi_F(S)$. By Lemma 3.3, $\operatorname{ord} S = 0$, i.e. $S = g(x, y, z) \in \mathbb{C}[x, y, z]$, and then $x = g(F_1, F_2, F_3)$. Evaluating at the first two collision points of (3), which share their F-image but have x-coordinates 1 and -1, gives $1 = -1$; contradiction. Hence $x \notin \operatorname{im}\Psi_F$.

For $n > 3$, write $W_n = W_3 \otimes W_{n-3}$ and set $\Psi = \Psi_F \otimes \mathrm{id}$. It is a unital endomorphism, injective by simplicity of W_n. The order filtration argument goes through verbatim: $\operatorname{gr}\Psi$ is the pullback along $\Phi \times \mathrm{id}_{\mathbb{C}^{2(n-3)}}$, which is dominant, so Ψ is strictly filtered; a preimage of $x \in W_n$ would give $x = g(F_1, F_2, F_3, x_4, \dots, x_n)$, and evaluation at the two collision points (with the remaining coordinates fixed) again gives $1 = -1$. □

Remark 3.4 (Mayner's finer results). Much more is true, and was established first in [May26]: the image of Ψ_F is exactly $\bigoplus_{\alpha\in\mathbb{N}^3} \mathbb{C}[F_1, F_2, F_3]\, D^\alpha$; none of $x, y, z, \partial_1, \partial_2, \partial_3$ lies in the image; the codimension of the image is tabulated in filtration degrees $d \le 7$; and the cokernel is not finitely generated. Note that the restriction of Ψ_F to the order-zero subalgebra $\mathbb{C}[x,y,z] \subset W_3$ is precisely the pullback $F^*\colon g \mapsto g \circ F$ — the classical non-surjective shadow. The mechanism "Ψ_F automorphism $\Rightarrow$ F invertible" is classical; see [vdE00] and [BKK07, p. 2]. Lemmas 3.2–3.3 are our packaging of that argument for this specific F.

4. The symplectic counterexample and the Poisson conjecture

Equip $\mathcal{O}(\mathbb{C}^6) = \mathbb{C}[x,y,z,p_1,p_2,p_3]$ with the standard Poisson bracket $\{g,h\} = \sum_i \big(\partial_{p_i} g\, \partial_{q_i} h - \partial_{q_i} g\, \partial_{p_i} h\big)$ (sign conventions are immaterial below). A polynomial map $\Phi\colon \mathbb{C}^6 \to \mathbb{C}^6$ satisfies $\{g \circ \Phi, h \circ \Phi\} = \{g,h\} \circ \Phi$ for all g,h if and only if its Jacobian satisfies $M^{\mathsf{T}} \Omega M = \Omega$ pointwise; in that case Φ^* is a unital Poisson-algebra endomorphism of P_3. The Poisson conjecture PC_n, in the formulation of [AvdE07], asserts that every such endomorphism of P_n is an automorphism, equivalently that every polynomial map of $\mathbb{C}^{2n}$ preserving the standard symplectic form is a polynomial automorphism.

Proof of Theorem B. The identity $M^{\mathsf{T}}\Omega M = \Omega$, the value $\det M = 1$, the component degrees, and the collision $\Phi(a_1) = \Phi(a_2) = \Phi(a_3) = (-\frac{1}{4}, 0, 0; 1, 2, 3)$ are exact symbolic facts (Proposition 2.3 and `dixmier_symplectic_verify.py`; the momenta of the a_i are $JF(q_i)^{\mathsf{T}}(1,2,3)^{\mathsf{T}}$). Hence Φ^* is a Poisson endomorphism of P_3; it is injective because Φ is dominant. It is not surjective: if $x = g \circ \Phi$ for some polynomial g, evaluation at a_1, a_2 (equal images, x-coordinates 1 and -1) gives $1 = -1$. So Φ^* is a proper endomorphism and PC_3 fails. The map Φ is generically 3:1 because its fibers biject with those of F (Section 2) and F is generically 3:1 (Proposition 5.3 below).

For $n > 3$, take $\Phi_n = \Phi \times \mathrm{id}_{\mathbb{C}^{2(n-3)}}$, arranging the extra coordinates into symplectic pairs; then $M_n^{\mathsf{T}} \Omega_{2n} M_n = \Omega_{2n}$, and non-surjectivity of Φ_n^* follows by the same two-line evaluation. Hence PC_n fails for all $n \geq 3$. □

Remark 4.1 (Relation to the formal implication, stated exactly). Given the failure of DC_3 (Theorem A, Mayner's), the failure of PC_3 follows abstractly from $\mathrm{PC}_3 \Rightarrow \mathrm{DC}_3$ [AvdE07], and is already asserted in secondary sources. The map Φ is Mayner's [May26, §8], as are the three properties verified in Proposition 2.3; his priority section lists the symplectic lift explicitly. What is added by Theorem B is the identification of Φ^* with the Poisson-endomorphism side of [AvdE07] — i.e. the label PC_n — together with the component degrees, the explicit momenta of the collision, and a non-invertibility proof that is two lines from (3), with no filtration and no reduction argument. Mayner does not use the words "Poisson conjecture" anywhere, and we have found no source that does in this connection; but the reader should treat Theorem B as a labelling and packaging contribution, not as the discovery of a new object.

Proof of Theorem C(ii). This is Lemma 3.2. (Part (i) is proved in Section 5.2.) □

Remark 4.2 (Quantization does not restore invertibility). Theorem C(ii) says the diagram one hopes for — classical symplectic maps below, Weyl-algebra maps above, symbol map connecting them — is realized exactly here: Ψ_F is a filtered quantization of Φ, with $\operatorname{gr} \Psi_F = \Phi^*$, and both are proper endomorphisms. Whatever mechanism produces the failure of injectivity of F survives passage to the Weyl algebra. This bears on the lifting program of [KBEY18] (and related work): a correct lifting theory must map the non-invertible symplectic endomorphism Φ to a non-invertible object, as the canonical quantization does. The Belov-Kanel–Kontsevich automorphism conjecture [BKK05] concerns groups of automorphisms and is untouched by these examples; see Section 1.5.

5. Equivariant structure: reduction, moment map, cover, image

5.1. The $\mathbb{C}^\times$-reduction and the master equation. Give (x,y,z) the weights $(1,-1,-2)$. The invariants of the action are generated by

$$u = 1 + xy, \qquad s = x^2 z.$$

The components of F are semi-invariant of weights $(-2,-1,1)$: reading them against the fiber coordinate x,

$$x^2 F_1 = A(u,s), \qquad xF_2 = B(u,s), \qquad F_3 = x\,C(u,s), \tag{4}$$

with

$$A = u^3 s + u(u-1)^2(3u+1), \qquad B = 3u^2 s + 9u^3 - 15u^2 + 4u + 2, \qquad C = 5 - 3u - s. \tag{5}$$

The invariant combinations of the targets, $p = 1 + F_2F_3$ and $q = F_1F_3^2$, define the *downstairs plane map*

$$G(u,s) = \big(1 + BC,\ AC^2\big). \tag{6}$$

Proposition 5.1 (Master equation; *Shaska* [Sha26, Thm. 8.3], obtained independently and machine-verified here)**.** *Let $k \geq 1$, $s = x^k z$, and let $A, B, C \in \mathbb{C}[u,s]$ be arbitrary. With $\mathcal{J}(f,g) = f_u g_s - f_s g_u$ and $G = (1 + BC,\ AC^k)$,*

$$\det JG \ = \ C^k\,\mathcal{B}_k, \qquad \mathcal{B}_k := C\,\mathcal{J}(B,A) + k\,A\,\mathcal{J}(B,C) + B\,\mathcal{J}(C,A), \tag{7}$$

identically in $\mathbb{C}[u,s]$. If moreover $F = (A/x^k,\ B/x,\ xC)$ is a polynomial map, then $\det JF = -\mathcal{B}_k$ *identically. Hence*

$$F \text{ is Keller} \iff \mathcal{B}_k \in \mathbb{C}^\times.$$

For the map (2) ($k=2$): $\det JG = 2C^2$, $\mathcal{B}_2 = 2$, *and* $\det JF = -2$.

The identity (7) isolates the design pattern: the Jacobian of the plane map G "parks" a full square C^k (for $k = 2$, the square C^2 on the line $\ell : 3u + s = 5$), and the fiber scaling $x \mapsto xC$ cancels it exactly. This divisorial form of the Keller condition — $\det JG = \kappa C^k$, i.e. order-k vanishing on the contracted locus with constant leading coefficient — is [Sha26, Thm. 7.1] (Thm. 6.1 of the July 22 version), and the expanded trilinear identity (7) itself is [Sha26, Thm. 8.3]; his Remark 8.4 identifies the exponent $k = \sum_{i\geq 2} r_i - 1$ in general. We obtained (7) independently and give the machine verification, but the result is his. Two further verified facts complete the picture:

- *G contracts the critical line $\ell = \{C = 0\}$* to the single point $(p,q) = (1,0)$; restricted to ℓ, $(A,B)|_\ell = (u^2 + u,\ 4u + 2)$ and the one-variable Wronskian $b\,a' - 2a\,b'$ equals 2 — the same constant as the Jacobian. The construction is tight along its critical line.
- The triple collision (3) sits exactly over the contraction point $(1,0)$: the three fiber points have (u,s)-invariants $u = 1$ (the $x = 0$ sheet) and $u = -\frac{1}{2}$, $s = \frac{13}{2}$ (twice, the two points on $\{C = 0\}$).

A heuristic for "why not $\mathbb{C}^2$": the mechanism needs a fiber direction to absorb the parked square; in the plane there is none (and Moh [Moh83] proved the planar conjecture through degree 100).

5.2. **The Hamiltonian lift and the moment map (new).**

Proof of Theorem C(i). Equivariance of F is the statement (4): $F(t \cdot q) = S_t F(q)$ with $S_t = \operatorname{diag}(t^{-2}, t^{-1}, t)$. Differentiating, $JF(t \cdot q)\, T_t = S_t\, JF(q)$ with $T_t = \operatorname{diag}(t, t^{-1}, t^{-2})$, whence $(JF(t \cdot q))^{-\mathsf{T}} = S_t^{-1}(JF(q))^{-\mathsf{T}} T_t$. Since the cotangent lift acts on momenta by T_t^{-1} upstairs and S_t^{-1} on target momenta, Φ intertwines the two lifted actions. The lifted actions are Hamiltonian with the quadratic moment maps $\mu = xp_1 - yp_2 - 2zp_3$ and $\nu = -2Q_1P_1 - Q_2P_2 + Q_3P_3$ (weights times q_ip_i), and the exact identity

$$\nu \circ \Phi = \mu$$

is verified symbolically in `dixmier_symplectic_verify.py`. □

Remark 5.2 (The parked square on the symplectic quotient). Informally, Theorem C(i) locates the mechanism of Section 5.1 inside symplectic geometry: Φ descends to the symplectic quotients of the μ- and ν-level sets, where its reduced Jacobian degenerates along (the image of) the critical line $\{C = 0\}$ — the parked square lives on the quotient — while the group direction, the fiber coordinate x of (4), carries the compensating factor that keeps $\det M \equiv 1$ upstairs. We use this only as an organizing picture; the verified content is the intertwining identity and the master equation.

5.3. **Arithmetic of the** 3:1 **cover.**

Proposition 5.3 (machine-verified; *the conclusions of (i) and (ii) are anticipated*, see the note after the proposition)**.** *Eliminating s from $G = (p, q)$, the minimal polynomial of u over $\mathbb{C}(p, q)$ is the cubic*

$$\Delta_1\, u^3 + \big((p-1)^2 - 12q\big)\, u - 4q = 0, \tag{8}$$

where

$$\Delta_1 = p^3 - 4p^2 - 18pq + 5p + 27q^2 + 34q - 2.$$

Moreover:

- (i) *(Trace identity.) The cubic has no u^2 term: the three preimages of a generic point satisfy $(1 + x_1y_1) + (1 + x_2y_2) + (1 + x_3y_3) = 0$. On the collision fiber: $1 - \frac{1}{2} - \frac{1}{2} = 0$.*
- (ii) *(Discriminant.)* disc $= -4\,\Delta_1\Delta_2^2$ *with $\Delta_2 = p^3 - 3p^2 - 18pq + 3p + 54q^2 + 18q - 1$, and Δ_1 is irreducible over $\mathbb{Q}$ — in particular squarefree of odd degree 3 in p. Hence* disc *is not a square in $\mathbb{C}(p, q)$: the Galois group of (8) over $\mathbb{C}(p, q)$ is the full symmetric group S_3, and the cover is not Galois (see Remark 5.5 for the descent to $\mathbb{C}(p, q)$).*
- (iii) *(3:1.) s is a rational function of (u, p, q) (via the degree-one subresultant, an identity verified exactly), so $[\mathbb{C}(u, s) : \mathbb{C}(p, q)] = 3$ and F is generically 3-to-1.*

(Verified in `cover_verify.py`.)

Remark 5.4 (Attribution for Proposition 5.3). Only the coordinate is ours. That the function-field extension has degree exactly 3 with full S_3 monodromy, that the discriminant is not a square, and that the deck group is therefore trivial, were all established on July 20, 2026 in [MO26] (with an explicit cubic model in the weight-(-1) coordinate $t = y + 1/x$, discriminant $-4Q$ with $Q = 27a^2c^2 - 18abc + 16a + b^3c - b^2$, and the Campbell/Razar/Wright Galois-case theorem invoked for precisely the reason we invoke Campbell's below), and independently by Lou [Lou26] the same day; Mayner [May26, §4.4] has the same S_3 statement with his discriminant factorization $-4S^2\Delta$, and Shaska [Sha26, Thm. 4.4(2)] states it in the s-coordinate with $\Delta(P, Q) = -4P^3 + 4P^2 + 72PQ - 64Q - 108Q^2$. The trace identity (i) is the same phenomenon as Mayner's "the x-coordinates of the preimages sum to zero" [May26, §4.2] and

Shaska's "the three roots always sum to $2 = |\det JF|$" [Sha26, Rem. 5.4], transported to the invariant coordinate u; the three cubics are genuinely different cubics (different primitive elements of the same cubic extension), and we do not know whether the trace-free shape in u specifically is forced.

Remark 5.5 (From the $\mathbb{Q}$-certificates to $\mathbb{C}(p,q)$). The certificates behind Proposition 5.3 operate over $\mathbb{Q}$, and irreducibility over $\mathbb{Q}(p,q)$ does not by itself preclude a factorization over $\mathbb{C}(p,q)$; the gap closes as follows. F is an everywhere-local biholomorphism (Proposition 2.1(i)), so each of the three distinct points of the collision fiber (3) has a neighborhood mapped biholomorphically onto a neighborhood of $(-\frac{1}{4}, 0, 0)$; every nonempty Zariski-open subset of $\mathbb{C}^3$ meets the intersection of these image neighborhoods, so some target with $t \neq 0$ and $(p,q) \neq (1,0)$ has at least three F-preimages. Its F-fiber is in bijection with the G-fiber over (p,q) (Section 5.4), whose points all satisfy $C \neq 0$, where $\det JG = 2C^2 \neq 0$; so G is a local biholomorphism at each of the ≥ 3 fiber points, and the same openness argument, now downstairs, produces a Zariski-generic point of $\mathbb{C}^2$ with at least three G-preimages. Since the generic fiber count of a dominant map between irreducible varieties of equal dimension equals the function-field degree in characteristic zero, $[\mathbb{C}(u,s) : \mathbb{C}(p,q)] \geq 3$. On the other hand $s \in \mathbb{C}(u,p,q)$ by (iii), so $\mathbb{C}(u,s) = \mathbb{C}(p,q)(u)$, and u satisfies the cubic (8): the degree is ≤ 3. Hence the degree is exactly 3, the cubic is the minimal polynomial of u over $\mathbb{C}(p,q)$ — in particular irreducible over $\mathbb{C}(p,q)$, with no descent argument needed — and the monodromy group is transitive. Finally, disc $= -4\Delta_1\Delta_2^2$ is a square in $\mathbb{C}(p,q)$ if and only if $-\Delta_1$ is; because $\mathbb{C}[p,q]$ is a UFD, that would force $-\Delta_1$ to be the square of a polynomial, which is impossible: Δ_1 is squarefree — squarefreeness is the condition $\gcd(\Delta_1, \partial_p\Delta_1) = 1$, insensitive to extending the ground field in characteristic zero — of odd degree 3 in p, while nonconstant polynomial squares have even degrees. So disc is not a square in $\mathbb{C}(p,q)$, the monodromy is not contained in A_3, and, being transitive, it is all of S_3.

Remark 5.6 (Campbell's theorem makes S_3 a necessity). By a theorem of Campbell [Cam73] — proved independently, in general characteristic zero, by Razar and by Wright — a Keller map whose extension of rational function fields is Galois is invertible. A counterexample of covering degree 3 therefore *had* to be non-Galois, i.e. to have monodromy S_3 rather than $\mathbb{Z}/3$, and Proposition 5.3(ii) confirms that Alpöge's map threads exactly that needle. This observation is also not ours: it is made in [MO26] on July 20, 2026, with the same citation. Whether the trace identity (i) is forced by the mechanism or is extra tuning is an open question worth testing on any future examples with other weights.

5.4. **The image theorem.** The statement of Theorem D is not new; see the priority note in its statement and Section 1.4. The proof below is our own and is included because the G-fiber correspondence it sets up is reused in Theorem E, and because the transfer $\operatorname{im}\Phi = \mathbb{C}^6 \setminus (\Gamma \times \mathbb{C}^3)$ needs it. Readers who prefer the published treatments should consult [Ula26, Thm. 4.2], which routes the fiber count through a binary cubic $2pS^3 - qS^2T + 2ST^2 - rT^3$ and an incidence variety in $\mathbb{C}^3 \times \mathbb{P}^1$, or [Sha26, Prop. 5.1], which routes it through the contracted line.

Proof of Theorem D. Write the target coordinates as (X, Y, t).

The plane $t = 0$ is covered. On the sheet $x = 0$ the map is $F(0,y,z) = (z + 4y^2,\ y,\ 0)$, visibly bijective onto the plane $\{t = 0\}$.

Targets with $t \neq 0$. By (4), a preimage with $F_3 = t \neq 0$ has $x \neq 0$ and $C(u,s) \neq 0$, and corresponds to a point of the G-fiber over

$$(p,q) = (1 + tY,\ t^2X),$$

with $x = t/C(u,s)$ recovered uniquely; conversely every G-fiber point with $C \neq 0$ arises this way. If $(p,q) \neq (1,0)$, every G-fiber point automatically has $C \neq 0$ (since $C = 0$ forces $(p,q) = (1,0)$), so the F-fiber over (X,Y,t) is in bijection with the G-fiber over (p,q). The axis targets $(0,0,t)$, $t \neq 0$ — exactly the case $(p,q) = (1,0)$ — are hit exactly once, by $(t/2,0,0)$: the system $A = B = 0$, $C \neq 0$ has the single solution $(u,s) = (1,0)$, where $C = 2$ (reduced Gröbner basis over $\mathbb{Q}$, asserted in `min_verify.py` part `axis`), whence $x = t/2$, $y = 0$, $z = 0$.

The line $q = 0$ is covered. On the $u = 0$ sheet $q \equiv 0$ and $p = 11 - 2s$ takes every value; so every (p,q) with $q = 0$ has a G-fiber point, and roots $u_0 = 0$ of the cubic — which occur only when the constant term $-4q$ vanishes — never need to be lifted.

From cubic roots to fiber points. Fix a target (p,q) with $q \neq 0$ and let u_0 be any root of the cubic (8). Viewed as polynomials in s, $p - P$ and $q - Q$ have degrees 2 and 3 with s-leading coefficients $-3u^2$ and u^3 respectively (from BC and AC^2), and the cubic (8) is an irreducible factor of their s-resultant — indeed the resultant equals $-u^3$ times (8), the extraneous factor being supported on $\{u = 0\}$ (`cover_verify.py` extracts (8) from precisely this resultant). Since $q \neq 0$, the constant term $-4q$ of the cubic is nonzero, so $u_0 \neq 0$ and both s-leading coefficients $-3u_0^2$, u_0^3 are nonzero; hence specializing $u = u_0$ commutes with taking the resultant, and the specialized resultant vanishes because the cubic does. Two univariate polynomials with nonvanishing leading coefficients and vanishing resultant have a common root: there is $s_0 \in \mathbb{C}$ with $(u_0, s_0) \in G^{-1}(p,q)$. So the G-fiber over any (p,q) with $q \neq 0$ is nonempty whenever the cubic has a root.

Emptiness locus of G. By the previous two steps, the fiber can be empty only where the cubic has no root at all: only where its leading coefficient Δ_1 and its linear coefficient $(p-1)^2 - 12q$ both vanish while the constant term $-4q$ does not. Solving, the unique such point is $(p,q) = (\frac{7}{3}, \frac{4}{27})$, and a Gröbner certificate over $\mathbb{Q}$ shows the ideal generated by the G-equations at that point is the *unit ideal*: that fiber is indeed empty (Nullstellensatz; `min_verify.py`, `cover_verify.py`).

The missed curve. Pulling $(p,q) = (\frac{7}{3}, \frac{4}{27})$ back through $(p,q) = (1 + tY, t^2X)$ gives, for each $t \neq 0$, the unique missed target $\left(\frac{4}{27t^2}, \frac{4}{3t}, t\right)$, i.e. the curve Γ. Every other target is attained, so $\operatorname{im} F = \mathbb{C}^3 \setminus \Gamma$, and $\operatorname{im} \Phi = \mathbb{C}^6 \setminus (\Gamma \times \mathbb{C}^3)$ by the fiber correspondence of Section 2.

Fiber counts. Generic fibers have three points. Over $\{\Delta_2 = 0\}$ (with $\Delta_1 \neq 0$) two sheets merge; over $\{\Delta_1 = 0\} \setminus \{(\frac{7}{3}, \frac{4}{27})\}$ the cubic drops degree and exactly one preimage remains — two sheets escape to infinity; at $(\frac{7}{3}, \frac{4}{27})$, i.e. over Γ, all three sheets have escaped and the fiber is empty. □

So F is an everywhere-local biholomorphism of $\mathbb{C}^3$, generically 3:1, surjective except for one explicit punctured rational curve — a picture worth keeping: over Γ the missing preimages have escaped to infinity along $\{\Delta_1 = 0\}$.

Remark 5.7 (Transport to Ψ_F). At order zero, the deficiency of Ψ_F *is* the deficiency of F^*: $\mathbb{C}[F] \subsetneq \mathbb{C}[x,y,z]$, an inclusion of polynomial algebras dual to F itself, whose geometric failure of surjectivity is exactly the curve Γ. In higher filtration degrees, Mayner's computations [May26] (exact image $\bigoplus_\alpha \mathbb{C}[F]D^\alpha$, codimension table for $d \leq 7$, non-finitely-generated cokernel) quantify the deficiency of Ψ_F; at the symbol level Lemma 3.2 bounds the image by $\operatorname{im} \Phi^*$, whose geometric deficiency is $\Gamma \times \mathbb{C}^3$ (Theorem D). We expect the growth of coker Ψ_F to be governed by functions along Γ and the S_3 monodromy of Proposition 5.3, but we leave a precise statement to future work.

6. Degree minimality in the equivariant class

Fix the weights $(1,-1,-2)$, i.e. $k=2$, $u=1+xy$, $s=x^2z$, and consider all maps of the shape (4): $F=(A/x^2, B/x, xC)$ with $A,B,C\in\mathbb{C}[u,s]$. Writing $A=A_0(u)+A_1(u)s+A_2(u)s^2+\cdots$ and likewise for B,C, the condition that F be a polynomial map (*lift conditions*) is

$$(u-1)^2 \mid A_0, \qquad B_0(1)=0,$$

and by Proposition 5.1 the Keller condition is $\mathcal{B}_2=\lambda\in\mathbb{C}^\times$. We refer to this family as *the equivariant class*; the map (2) is the member (5) with $\lambda=2$.

6.1. **The degree dictionary and normalization.** The monomial substitution $u^is^j\mapsto x^{i+2j}y^iz^j$ is injective on exponent pairs, so there is no cancellation of leading terms and degrees upstairs are read off exactly. The condition $\deg F\le 6$ caps the s-expansion coefficients precisely as follows:

	s^0-part	s^1-part	s^2-part
$A:$	$\deg\le 4$	$\deg\le 2$	$\deg\le 1$
$B:$	$\deg\le 3$	$\deg\le 2$	constant
$C:$	$\deg\le 2$	$\deg\le 1$	—

Conjugating by the two diagonal $\mathbb{C}^\times$-automorphisms upstairs scales $(A,B,C)\mapsto(\nu r^{-2}A,\ \nu r^{-1}B,\ \nu rC)$ and $\lambda\mapsto\nu^3r^{-2}\lambda$; this legitimately normalizes $\lambda=1$ together with one further nonzero leading coefficient to 1 in each case below.

6.2. **The anchor lemma.**

Lemma 6.1 (Anchor; *Shaska* [Sha26, Lem. 8.5], obtained independently here)**.** *For every Keller map in the equivariant class, of any degree,*

$$\lambda=C_0(1)\cdot B_0'(1)\cdot A_1(1).$$

In particular every Keller lift — counterexample or automorphism — satisfies $C_0(1)\neq 0$, $B_0'(1)\neq 0$, and $A_1(1)\neq 0$.

Proof. Evaluate $\mathcal{B}_2 = C\,\mathcal{J}(B,A)+2A\,\mathcal{J}(B,C)+B\,\mathcal{J}(C,A)$ at $(u,s)=(1,0)$. The lift conditions give $A(1,0)=A_0(1)=0$ and $A_u(1,0)=A_0'(1)=0$ (double root), and $B(1,0)=B_0(1)=0$; so the second and third terms vanish, and in the first, $\mathcal{J}(B,A)(1,0)=B_uA_s-B_sA_u=B_0'(1)A_1(1)$. Multiplying by $C(1,0)=C_0(1)$ gives the identity. (Machine-verified at full $\deg\le 6$ generality in `min_verify.py`; the derivation above is degree-free.) For the map (5): $2=2\cdot 1\cdot 1$. ☐

Remark 6.2 (Attribution for Lemma 6.1)*.* Shaska states this as: evaluating the master equation at the base point O of the quotient plane gives $\Lambda(O)\,\mathcal{J}(B,A)(O)=\kappa$, whence $\Lambda(O)\neq 0$ and $dA\wedge dB|_O\neq 0$ [Sha26, Lem. 8.5]. Translating his base point to $(u,s)=(1,0)$ and using the lift conditions to kill $A_u(1,0)$ turns $\mathcal{J}(B,A)(O)=B_uA_s-B_sA_u$ into $B_0'(1)A_1(1)$, which is the displayed factorization: the two statements are the same lemma, and his is earlier (July 25, 2026, in v2 of [Sha26]). We keep our formulation because the factored form is what the case analysis of Theorem E consumes.

6.3. **The no-go lemma.** The following appears to be new; we have found no analogue in [May26, Sha26, Ula26, Lou26, MO26]. The nearest published statements are [Sha26, Rem. 8.8], which handles the case Λ constant (where the graded problem in dimension n collapses to the ordinary Jacobian conjecture in dimension $n-1$), and [Sha26, Thm. 10.10], which handles A and B both of degree one in the invariants. Lemma 6.3 constrains the s-degree rather than the total degree, and is what makes Branch I of Theorem E finite.

Lemma 6.3 (No-go; all weights $k \geq 1$; *new*)**.** *Fix $k \geq 1$ and weights $(1, -1, -k)$. If C is s-free and A, B are at most affine in s, then every Keller lift in the class is a polynomial automorphism of $\mathbb{C}^3$ — at any degree.*

Proof. Write $A = A_0(u) + A_1(u)s$, $B = B_0(u) + B_1(u)s$, $C = C_0(u)$, and recall the invariant outputs $p - 1 = BC$ and $q = AC^k$. Note first that $C_0 \neq 0$ as a polynomial (else $\mathcal{B}_k = 0$), and that if $A_1 = B_1 = 0$ then $\mathcal{B}_k = 0$ and the map is not Keller; so at least one of A_1, B_1 is nonzero. Throughout, (X, Y, T) denote the target coordinates of F, so that $p = 1 + YT$ and $q = XT^k$ are rational in the target.

Main case $A_1B_1 \neq 0$. The bracket $\mathcal{B}_k$ is affine in s; setting its s-coefficient to zero and dividing by $A_1B_1C_0$ gives $B_1'/B_1 - A_1'/A_1 - (k-1)\,C_0'/C_0 = 0$, i.e. $B_1 = \kappa A_1 C_0^{k-1}$ for a constant $\kappa \in \mathbb{C}^\times$ (a logarithmic-derivative computation). With this substitution the bracket collapses to the verified identity

$$\mathcal{B}_k = A_1 \cdot \big(C_0B_0 - \kappa\, C_0^k A_0\big)'.$$

A polynomial times a derivative being the nonzero constant λ forces $A_1 \in \mathbb{C}^\times$ and $C_0B_0 - \kappa C_0^k A_0 = \tilde{\lambda}u + \mu$ with $\tilde{\lambda} = \lambda/A_1 \neq 0$. A second verified identity says that, with these forced shapes, the s-terms of $(p-1) - \kappa q$ cancel *identically*:

$$(p-1) - \kappa q \;=\; C_0B_0 - \kappa\, C_0^k A_0 \;=\; \tilde{\lambda}u + \mu.$$

Now invert F rationally, in three steps. First, $u = \big((p-1) - \kappa q - \mu\big)/\tilde{\lambda}$ is a polynomial in (p, q), hence rational in the target. Second, at fixed u the relation $p - 1 = (B_0 + B_1 s)\, C_0$ is affine in s with s-coefficient $B_1C_0 = \kappa A_1 C_0^k \neq 0$ in $\mathbb{C}(u)$, so $s = \big((p-1) - B_0C_0\big)/\big(\kappa A_1 C_0^k\big)$ is rational in the target. Third, $x = F_3/C(u,s) = T/C_0(u)$, then $y = (u-1)/x$ and $z = s/x^k$, are rational in the target. So F is birational, and a birational Keller map is a polynomial automorphism [vdE00, Corollary 1.1.35] (equivalently [BCW82, Theorem 2.1]; the birational case goes back to Keller [Kel39]).

Edge case $B_1 = 0$, $A_1 \neq 0$. Here the bracket collapses to $\mathcal{B}_k = A_1\,(C_0B_0)'$ — the $\kappa = 0$ instance of the displayed identity — so again $A_1 \in \mathbb{C}^\times$ and $C_0B_0 = \tilde{\lambda}u + \mu$ with $\tilde{\lambda} \neq 0$. Since B is s-free, $p - 1 = B_0C_0 = \tilde{\lambda}u + \mu$, so u is affine in p; then $q = (A_0 + A_1 s)\, C_0^k$ gives $s = \big(q\, C_0^{-k} - A_0\big)/A_1$, rational in the target, and x, y, z are recovered as above: F is birational, hence an automorphism.

Edge case $A_1 = 0$, $B_1 \neq 0$. Here the bracket reduces to the relation $B_1\,(C_0^k A_0)' = -\lambda\, C_0^{k-1}$. Compare orders of vanishing at $u = 1$, writing $a = \operatorname{ord}_1 A_0$ and $c = \operatorname{ord}_1 C_0$: the lift condition $(u-1)^k \mid A_0$ gives $a \geq k$, so the left side has $\operatorname{ord}_1 = \operatorname{ord}_1 B_1 + kc + a - 1$ (the derivative drops the order by exactly one in characteristic zero), while the right side has $\operatorname{ord}_1 = (k-1)c$; equality forces $\operatorname{ord}_1 B_1 + c + a = 1$, impossible for $k \geq 2$ since $a \geq k \geq 2$ — this subcase is empty, consistent with Theorem E. For $k = 1$ the relation reads $B_1\,(C_0A_0)' = -\lambda$: a product of polynomials equal to a nonzero constant, so $B_1 \in \mathbb{C}^\times$ and C_0A_0 is affine in u with nonzero slope. Then $q = AC = A_0C_0$ is affine in u with nonzero slope: u is affine in q; then $p - 1 = (B_0 + B_1 s)C_0$ recovers s rationally ($B_1C_0 \neq 0$), and x, y, z follow as above: F is birational, hence an automorphism. *(The two identities displayed in the main case are verified in `core_verify.py`.)* □

Corollary 6.4. *A counterexample in the class must either put s inside C, or put s^2-terms inside A or B. Alpöge's map takes the first exit: $C = 5 - 3u - s$.*

6.4. **The minimality theorem.**

Proof of Theorem E. By the degree dictionary, $\deg F \leq 6$ means the caps in the table above. We split on the shape of C.

Branch I: C is s-free ($C = C_0(u)$, $\deg C_0 \leq 2$).

- If C_0 is constant, then G is an affine map composed with a planar Keller pair of degree ≤ 4, which is invertible by Moh's theorem [Moh83]; the lift is an automorphism.
- If A, B are at most affine in s, Lemma 6.3 applies: the lift is an automorphism.
- There remains the s^2-*sector*: C_0 nonconstant and $(A_2, B_2) \neq (0, 0)$. After normalization this is six polynomial systems in the coefficient unknowns (nonvanishing of the leading coefficient of C_0 imposed by a Rabinowitsch variable). For all six, msolve returns the *unit ideal* over $\mathbb{Q}$: by the Nullstellensatz the sector is empty — it contains no Keller map whatsoever.

Branch II: $C_1 \neq 0$ (C genuinely involves s). Two systems at *full* generality — all coefficients of A (through s^2, degrees capped), B (through s^2), C_0, and C_1 free, with only the legitimate normalizations $\lambda = 1$ and C_1 monic (respectively constant $= 1$); no hand-derived reduction is load-bearing. Both return the unit ideal over $\mathbb{Q}$: nothing exists in this sector below degree 7, automorphisms included.

The eight systems are:

Leaf	Sector	Equations	Unknowns	Result ($\mathbb{Q}$ and $\mathbb{F}_{32003}$)
II-f1	C_1 monic linear	26	19	(1)
II-f0	$C_1 = 1$	23	18	(1)
I-B2-g2	$B_2 = 1$, $\deg C_0 = 2$	23	18	(1)
I-B2-g1	$B_2 = 1$, $\deg C_0 = 1$	19	17	(1)
I-A2L-g2	A_2 monic linear, $\deg C_0 = 2$	20	17	(1)
I-A2L-g1	A_2 monic linear, $\deg C_0 = 1$	17	16	(1)
I-A2c-g2	$A_2 = 1$, $\deg C_0 = 2$	19	16	(1)
I-A2c-g1	$A_2 = 1$, $\deg C_0 = 1$	16	15	(1)

Positive control. At the degree-7 caps, the torus-scaled map (5) satisfies the identical normalized system ($\mathcal{B}_2 = 1$, C_1-coefficient $= 1$) — verified exactly by substitution: with $\nu = 2^{-1/5}$ (exact, $\nu^5 = \frac{1}{2}$) and $r = -1/\nu$, the scaled triple ($\nu r^{-2} A$, $\nu r^{-1} B$, $\nu r C$) has C_1-coefficient 1 and bracket identically 1; the asserted form of this check is `min_verify.py` part `d7control`. (An msolve reduced-basis run on the degree-7 control system did not terminate within a 600-second cap — expected for a nonempty positive-dimensional variety; the exact substitution above is the control, and the computational record is `certificates/D7CONTROL-NEGATIVE-RESULT.md` in the repository.) The pipeline provably contains the counterexample at degree 7 and provably contains nothing at degree ≤ 6.

All eight emptiness certificates were computed over $\mathbb{Q}$ (msolve multi-modular F4); the mod-32003 reproductions were run independently in both msolve and SymPy's Gröbner engine (`min_verify.py` parts `I` and `II`). SymPy additionally verifies the structural layer identities (Lemma 6.1 and the layer factorizations) used to *organize*, but not to prove, the case split. □

Remark 6.5 (Anatomy of the exit at degree 7). In the sector $A_2 = B_2 = 0$, $C_1 \neq 0$, a layer computation shows the top s-layer of the bracket forces $B_1^3 = \rho A_1^2 C_1$ for a constant ρ, i.e. $3 \deg B_1 = 2 \deg A_1 + \deg C_1$. Under the deg ≤ 6 caps only the patterns $(0, 0, 0)$ and $(1, 1, 1)$ survive, and both die against Lemma 6.1; the example lives on the pattern $(\deg B_1, \deg A_1, \deg C_1) = (2, 3, 0)$ — $6 = 6 + 0$ — which forces $\deg F_1 = 7$. Degree 7 is not an accident; it is the first rung the mechanism can reach.

This layer relation is a special case of a published one. Shaska [Sha26, Thm. 10.6] shows that the leading forms $\hat{A}, \hat{B}, \hat{\Lambda}$ of any graded Keller map are, up to constants, products of powers of a common set of linear forms, subject to the Diophantine system $\beta_i + r\gamma_i = (\nu_B/e)m_i$, $\alpha_i + s\gamma_i = (\nu_A/e)m_i$; and his Example 10.9 works out that system at the degrees $\mathbf{d} = (4,3,1)$ of Alpöge's map, finding $h = u(3u+v)$, $m = (1,1)$, $\beta = (2,1)$, $\gamma = (0,1)$, $\alpha = (3,1)$ and concluding that "the leading data of the announced counterexample is thus the solution of a small system of linear equations in nonnegative integers, and not an accident". Our relation $B_1^3 = \rho A_1^2 C_1$ is the same constraint read one s-layer at a time, and the conclusion above is the same conclusion; his is earlier (July 25, 2026) and more general. What the remark adds is the pairing of that constraint with the anchor identity to *exclude* the surviving low-degree patterns — which is exactly the content that Theorem E certifies and that [Sha26, Rem. 10.8] explicitly says the stratification alone does not give ("emptiness must therefore be proved stratum by stratum").

Remark 6.6 (Scope caveats, stated plainly). (i) Theorem E is relative to this symmetry class: weights $(1,-1,-2)$ and the lift shape (4). Other weights $(1,-1,-k)$ and non-equivariant maps are untouched; "no degree-≤ 6 counterexample in $\mathbb{C}^3$" remains open — unconditionally, invertibility of Keller maps is known only through degree 2, in every dimension, by Wang's theorem [Wan80]. (ii) The characteristic-zero emptiness certificates rest on msolve's multimodular F4 over $\mathbb{Q}$; the mod-p runs concur. An independent certificate checker that replays the unit-ideal cofactor identities has not yet been built, so a reader who does not trust msolve over $\mathbb{Q}$ has at present only the mod-32003 cross-reproduction. (iii) Moh's theorem and Campbell's theorem are quoted from the literature. (iv) Certificates for the weights $k = 1$ and $k = 3$ are in preparation and are not claimed here. (v) Emptiness *inside* a class is not emptiness outside it: ulam.ai [Ula26, Thm. 5.1, Cor. 5.3] produces a (λ, a, c, H)-family with the same cubic geometry, and nonproper Keller maps of every generic degree $d \geq 3$; Gao [Gao26] produces further explicit counterexamples in every dimension > 2 by a tangent-sweep construction; Migus [Mig26] determines the possible generic degrees over $\mathbb{R}$. None of these is a degree-≤ 6 member of our equivariant class, but they are a reminder that the map is not isolated in any broad sense — a point Mayner records as a correction to his own rigidity claim [May26, §12].

Remark 6.7 (Stable reductions). By Bass–Connell–Wright [BCW82], Yagzhev [Yag80], and Drużkowski [Dru83], the example yields cubic-homogeneous (even cubic-linear-form) counterexamples in some higher $\mathbb{C}^N$; in the discussion thread of [Spe26] an explicit cubic-homogeneous reduction in 24 variables was described (Thompson), and Long [Lon26] tracks a conservative Bass–Connell–Wright reduction of the announced map to 79 variables en route to the Gaussian Moments conjecture. Explicit degree/dimension bookkeeping, and the smallest such N, would be a worthwhile follow-up, as would the minimality question for the other weights $(1,-1,-k)$ — the master equation (7) is already general, and the same certificate pipeline applies.

Remark 6.8 (Other consequences drawn elsewhere). For orientation, the July 2026 consequence-drawing around this map includes: the Dixmier conjecture, Mayner [May26]; the Gaussian Moments conjecture, Long [Lon26]; the Hessian conjecture in five variables, Meng–Yang [MY26]; the separable Jacobian conjecture in characteristic 2, Huq-Kuruvilla [HK26]; quantum inequivalence of scalar field redefinitions, Zhu [Zhu26]; the structure of the space of Keller maps, Jelonek [Jel26]; real generic degrees, Migus [Mig26]; further families, ulam.ai [Ula26] and Gao [Gao26]; and the graded classification, Shaska [Sha26]. The present note contributes to none of those lines except the last.

7. Reproducibility

All artifacts sit beside this note. Trust model, stated exactly (it matches Remark 6.6(ii)): every polynomial identity in this note is an exact symbolic check asserted in SymPy 1.14 (arbitrary-precision rational arithmetic). The characteristic-zero emptiness certificates of Theorem E rest on msolve 0.10.1's certified multi-modular F4 over $\mathbb{Q}$; SymPy independently reproduces the mod-32003 runs, so the cross-system reproduction concerns the mod-p computations, while the characteristic-zero certificates rest on msolve alone. No numerical approximation occurs anywhere. A self-contained independent certificate checker (replaying the unit-ideal cofactor identities) is desirable and has not yet been built.

- `core_verify.py` — the map (2), $\det JF = -2$, the collisions, the collapse (4), the master equation (7) (all k, with symbolic function coefficients), the critical-line facts, and the two identities of Lemma 6.3.
- `cover_verify.py` — the cubic (8), trace identity, discriminant factorization, S_3 monodromy, rationality of s, and the image/escape analysis of Theorem D.
- `weyl_verify.py` — polynomiality of N, the CCR identities (A) and (B) of Proposition 3.1; writes `weyl_endomorphism.txt`, the nine fully expanded operator coefficients.
- `dixmier_symplectic_verify.py` — $M^{\mathsf{T}}\Omega M = \Omega$, $\det M = 1$, component degrees, the $\mathbb{C}^6$ triple collision, the moment-map identity $\nu \circ \Phi = \mu$, and an independent re-check of the CCR.
- `min_verify.py` — Lemma 6.1 at full deg ≤ 6 generality, the layer identities, the image certificate at $(\frac{7}{3}, \frac{4}{27})$ over $\mathbb{Q}$ (part `ident`); a SymPy Gröbner harness for the eight leaf systems mod 32003 (parts `I`, `II`); the degree-7 positive control by exact scaled substitution (part `d7control`); the upstairs identity $\det JF = -\mathcal{B}_k$ for $k = 1, 2, 3$ with free symbolic coefficients (part `kdet`); and the axis-target uniqueness certificate (part `axis`).
- `gen_ms.py`, `ms_*_c0.ms`, `ms_*_c32003.ms`, `out_*.txt` — msolve input files for the eight leaf systems in characteristic 0 and mod 32003, and the sixteen outputs (`out_*_q.txt` = characteristic 0, `out_*_p.txt` = mod 32003); every output is the reduced Gröbner basis [1].

```
python3 -m venv venv && venv/bin/pip install sympy
venv/bin/python core_verify.py
venv/bin/python cover_verify.py
venv/bin/python weyl_verify.py
venv/bin/python dixmier_symplectic_verify.py
venv/bin/python min_verify.py ident
venv/bin/python min_verify.py d7control
venv/bin/python min_verify.py kdet
venv/bin/python min_verify.py axis
venv/bin/python min_verify.py I    # SymPy GB, six Branch-I leaves, mod 32003
venv/bin/python min_verify.py II   # SymPy GB, two Branch-II leaves, mod 32003
brew install msolve
for f in ms_I-*_c0.ms ms_II-*_c0.ms; do b=${f#ms_}; \
  msolve -g 2 -f "$f" -o "out_${b%_c0.ms}_q.txt"; done          # char 0
for f in ms_I-*_c32003.ms ms_II-*_c32003.ms; do b=${f#ms_}; \
  msolve -g 2 -f "$f" -o "out_${b%_c32003.ms}_p.txt"; done      # mod 32003
```

Expected: every `out_*_q.txt` and `out_*_p.txt` contains the reduced Gröbner basis [1], matching the stored outputs.

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We are grateful to W. G. P. Mayner, whose note and report [May26] first put both the Weyl-algebra endomorphism *and* the symplectic cotangent lift on public record, and whose computations go beyond ours at several points; to T. Shaska [Sha26], whose graded framework contains, in greater generality and earlier, the master equation, the base-point identity, the divisorial form of the Keller condition, and the leading-form analysis that we had arrived at independently; to the anonymous author of [Ula26] for the earliest published fiber-count and image theorem; and to the participants of the Secret Blogging Seminar discussion [Spe26] — in particular the explicit 24-variable cubic-homogeneous reduction described there (Thompson). Discovering that a result one has proved was proved first by someone else is the ordinary condition of working on a problem that is three weeks old and crowded; we have tried to record it accurately rather than minimally. The debt to Levent Alpöge's example, and to Akhil Mathew for posing the problem, is total. Computations were carried out with Claude (Fable 5), SymPy 1.14, and msolve 0.10.1.

Note added (August 4, 2026). A final pre-release survey found two independent observations adjacent to Theorem C: a note of A. Husain (github.com/Cobord/Jacobian, August 1, 2026) constructs the cotangent-lift Poisson endomorphism and observes the intertwining of the weighted tori, posing the isotropy computation as open, without moment-map structure; and a blog comment of D. Fillmore (July 31, 2026, on Tao's digestion post) interprets the 3-to-1 fiber via Majorana spin addition, again without symplectic content. Neither states the moment-map identity $\nu \circ \Phi = \mu$ proved here; we record them for completeness.

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Independent researcher
Email address: `daniel@halfounce.io`